GOVERNMENT ENTERPRISE SPECIFICATION

02 Tech Stack & Framework Selection

Next.js vs Vite, Python FastAPI & Multi-Tier DBs • GovPulse Technical Architecture Suite • Version 1.0 (Production)

Technology Stack & Framework Selection Document

Project: GovPulse Autonomous Digital Government Platform

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1. Frontend Framework Decision: Next.js vs. React + Vite

For an enterprise-scale, mission-critical government platform serving millions of citizens alongside authorized high-security government officials, frontend architecture requires careful trade-off analysis.

Comparative Decision Matrix

ARCHITECTURAL CRITERION	NEXT.JS 14+ (APP ROUTER)	REACT + VITE (SINGLE PAGE APP)	WINNER & CRITICAL ANALYSIS FOR GOVPULSE
Zero-Trust Security & Token Isolation	**Exceptional (BFF Pattern):** Backend-for-Frontend (BFF) allows storing sensitive OAuth/OIDC tokens in HTTP-only, encrypted server session cookies.	**Moderate:** Tokens typically reside in browser `localStorage` or memory, making them vulnerable to Cross-Site Scripting (XSS) token theft.	**Winner: Next.js** (Mandatory for government-grade Zero-Trust compliance).

ARCHITECTURAL CRITERION	NEXT.JS 14+ (APP ROUTER)	REACT + VITE (SINGLE PAGE APP)	WINNER & CRITICAL ANALYSIS FOR GOVPULSE
	Client JavaScript never touches raw tokens.		
Initial Load Performance & SEO	**Instant (SSR / RSC):** Server-Side Rendering & Server Components deliver pre-rendered HTML with sub-second First Contentful Paint (FCP) on low-bandwidth 3G/4G citizen mobile devices.	**Slower Initial Load:** Heavy JS bundle must be downloaded and parsed before first render, leading to blank screen delays on low-end citizen devices.	**Winner: Next.js** (Crucial for universal citizen accessibility).
Official Officer Workspace UI	**Excellent:** Dynamic streaming SSR (`Suspense`), optimistic UI updates, and server actions for hardware token coordination.	**Excellent:** Rich client-side state manipulation, rapid hot-module reloading during development.	**Tie**
Deployment & Security Isolation	**Multi-Tier:** Can run in isolated Node.js container behind Envoy, serving as a hardened edge layer.	**Static CDN:** Deployable to static object storage/CDN. However, static assets require open public APIs.	**Winner: Next.js**
WebAuthn / FIDO2 Integration	Native browser WebAuthn API works seamlessly with Next.js server verification routes.	Works seamlessly via client-side WebAuthn API calling FastAPI endpoints.	**Tie**

Definitive Recommendation: Next.js 14+ (App Router) with Tailwind CSS

> **Decision Rationale:** Next.js provides the **Backend-for-Frontend (BFF)** security architecture essential for government Zero-Trust. It prevents tokens from leaking to the browser, accelerates citizen load times on mobile devices across remote

> regions via SSR, and enables unified API routing with automated security headers (`Strict-Transport-Security` , `Content-Security-Policy` , `X-Frame-Options: DENY`).

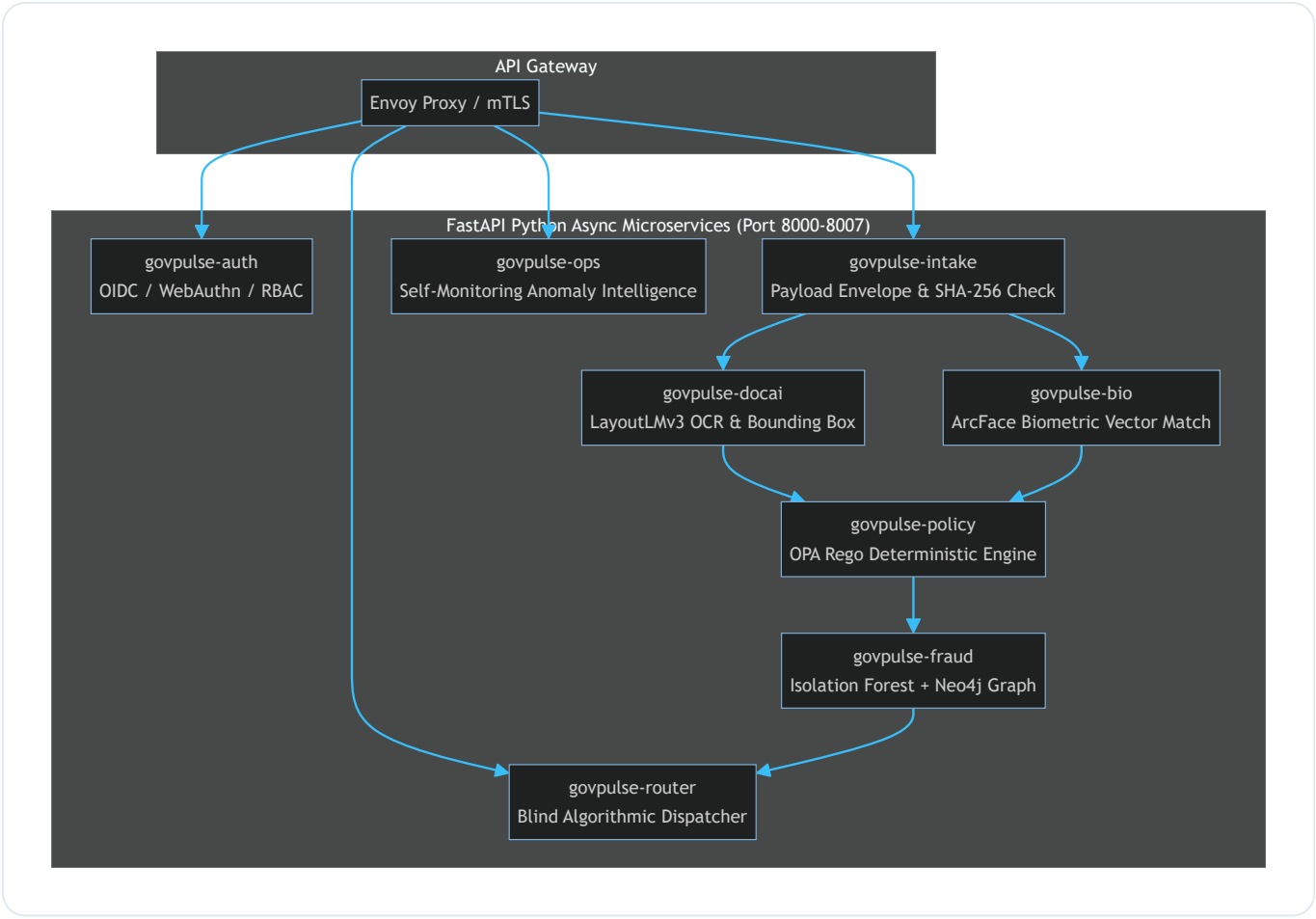
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2. Python Backend Microservices Architecture

The entire core processing, AI, policy, and workflow backend is built on **Python 3.12+**, leveraging its unparalleled ecosystem for asynchronous microservices, deep learning inference, cryptographic binding, and mathematical data pipelines.

PYTHON BACKEND ARCHITECTURE
• Framework: FastAPI (Async ASGI, OpenAPI 3.1, Sub-millisecond Pydantic v2 validati
• AI & Inference Engine: PyTorch 2.4+ / ONNX Runtime GPU (CUDA 12.4 acceleration)
• Document Vision: LayoutLMv3, PyMuPDF (fitz), PaddleOCR v4, OpenCV
• Biometric Matching: InsightFace / ArcFace (ResNet-100 backbone)
• Fraud ML & Stats: Scikit-Learn (Isolation Forest), NetworkX, PySpark
• Deterministic Policy: Python OPA Client (`open-policy-agent` / WASM runtime)
• Workflow Orchestration: Temporal Python SDK (Stateful Saga executions)
• Cryptography: `cryptography` (OpenSSL 3.3 bindings, ECDSA, Ed25519, AES-256-GCM)

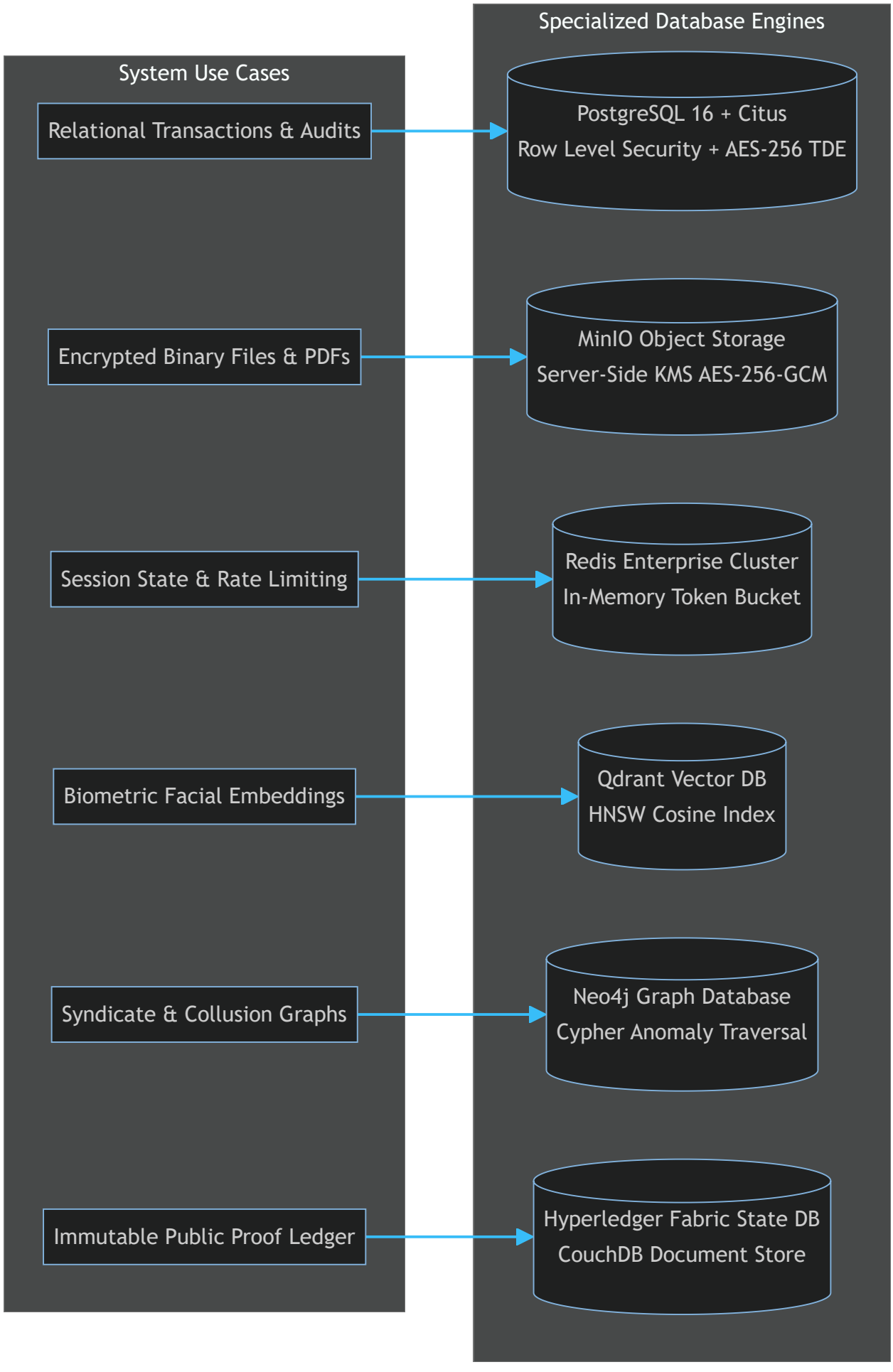
Microservices Decomposition:



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3. Multi-Tier Database Architecture & Security Evaluation

A national-scale digital government platform cannot use a single monolithic database. Different workloads require distinct storage guarantees for **performance, relational integrity, vector similarity, graph relationships, and cryptographic immutability**.


System Use Cases
Relational Transactions & Audits
Encrypted Binary Files & PDFs
Session State & Rate Limiting
Biometric Facial Embeddings
Syndicate & Collusion Graphs
Immutable Public Proof Ledger
Specialized Database Engines
PostgreSQL 16 + Citus
Row Level Security + AES-256 TDE
MinIO Object Storage
Server-Side KMS AES-256-GCM
Redis Enterprise Cluster
In-Memory Token Bucket
Qdrant Vector DB
HNSW Cosine Index
Neo4j Graph Database
Cypher Anomaly Traversal
Hyperledger Fabric State DB
CouchDB Document Store

Detailed Database Evaluation Matrix

DATABASE ENGINE	DESIGNATED SYSTEM ROLE	STORAGE MODEL	SECURITY & COMPLIANCE FEATURE
PostgreSQL 16 + Citus	Application state, citizen profiles, officer audit logs, SLA timers.	Relational SQL (ACID)	**Row-Level Security (RLS)** restricts data visibility per tenant/district; Transparent Data Encryption (TDE); immutable append-only audit tables.
MinIO Enterprise (S3 API)	Uploaded scanned certificates, tax forms, identity card binaries.	Object Store (Blob)	**Per-Object AES-256-GCM Envelope Encryption** with external HashiCorp Vault KMS; Object Locking (WORM: Write Once, Read Many).
Redis 7.4 Enterprise	Distributed session tokens, API rate-limiting, live SLA countdown cache.	In-Memory Key-Value	TLS encrypted communication, ACL authentication, sub-millisecond response for live SLA dashboards.
Qdrant Vector Database	512-dimensional facial biometric vector embeddings extracted via ArcFace.	HNSW Vector Index	High-dimensional cosine distance matching with hardware AVX-512 acceleration, payload filtering, and encrypted vector indices.
Neo4j Enterprise	Identity collision networks, address reuse, notary fraud syndicates.	Labeled Property Graph	Real-time Cypher graph queries detecting multi-hop collusions across unrelated applicants.
Hyperledger Fabric (CouchDB)	Block state database containing certificate hashes, issuer signatures, and status.	Distributed State Ledger	Cryptographically signed state transitions validated by multi-organization endorsement policies.

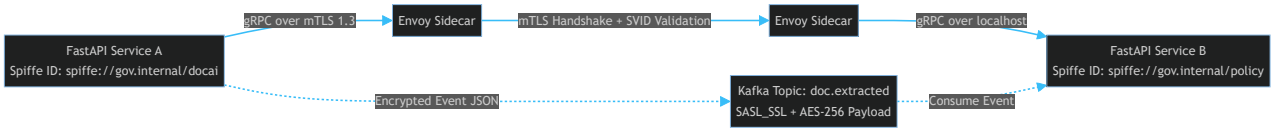
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4. Secure Inter-Service Communication Protocol

To prevent lateral movement inside the government cloud perimeter, all services adhere to **Zero-Trust Network Architecture (NIST SP 800-207)**:

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INTER-SERVICE SECURITY PROTOCOLS

1. Synchronous Control Plane: gRPC over mTLS (Mutual TLS 1.3 with SPIFFE/SPIRE IDs)
2. Asynchronous Data Plane: Apache Kafka with SASL/SCRAM authentication & TLS encry
3. Internal Service Authorization: Short-lived Spiffe JWT SVID tokens verified at p
4. Encryption at Rest: AES-256-GCM across all database volumes (LUKS / dm-crypt)
5. Key Management: HashiCorp Vault with Hardware Security Module (HSM) root of trus
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5. Complete Tech Stack Summary Table

CATEGORY	PRIMARY TECHNOLOGY	PURPOSE IN GOVPULSE
Frontend UI	**Next.js 14+ / React / Tailwind CSS**	Citizen application portal & single-officer review desk with SSR security.
Edge Security	**Cloudflare Enterprise + Envoy Proxy**	WAF, DDoS mitigation, mTLS, rate limiting, and ABAC gateway.
Core Backend	**Python 3.12+ (FastAPI)**	High-performance asynchronous REST and gRPC microservices.
Workflow & SLA	**Temporal.io (Python SDK)**	Distributed state machine enforcing 48-hour hard SLA escrow.

CATEGORY	PRIMARY TECHNOLOGY	PURPOSE IN GOVPULSE
AI Document Extraction	**LayoutLMv3 + PyMuPDF + PaddleOCR**	Multimodal OCR, key-value parsing, visual bounding boxes.
Biometric Vision	**InsightFace / ArcFace (PyTorch)**	Vectorized facial identity verification against national ID records.
Statutory Policy Engine	**Open Policy Agent (OPA) / Rego**	Deterministic legal eligibility evaluation with statutory code citations.
Fraud & Anomaly ML	**Isolation Forest + Neo4j Graph DB**	Document tampering anomaly scoring and identity syndicate detection.
Relational Storage	**PostgreSQL 16 + Citus**	Core application state with Row-Level Security (RLS).
Encrypted Object Store	**MinIO Enterprise (Vault KMS)**	Off-chain encrypted document binary storage (AES-256-GCM).
Vector Database	**Qdrant**	Biometric facial embedding similarity search.
Permissioned Blockchain	**Hyperledger Fabric v2.5**	Immutable certificate hash anchoring, public verification, and audit trail.
Hardware PKI / Sign	**WebCrypto / PKCS#11 / ECDSA P-256**	Officer digital signature committing issuance to the blockchain.